

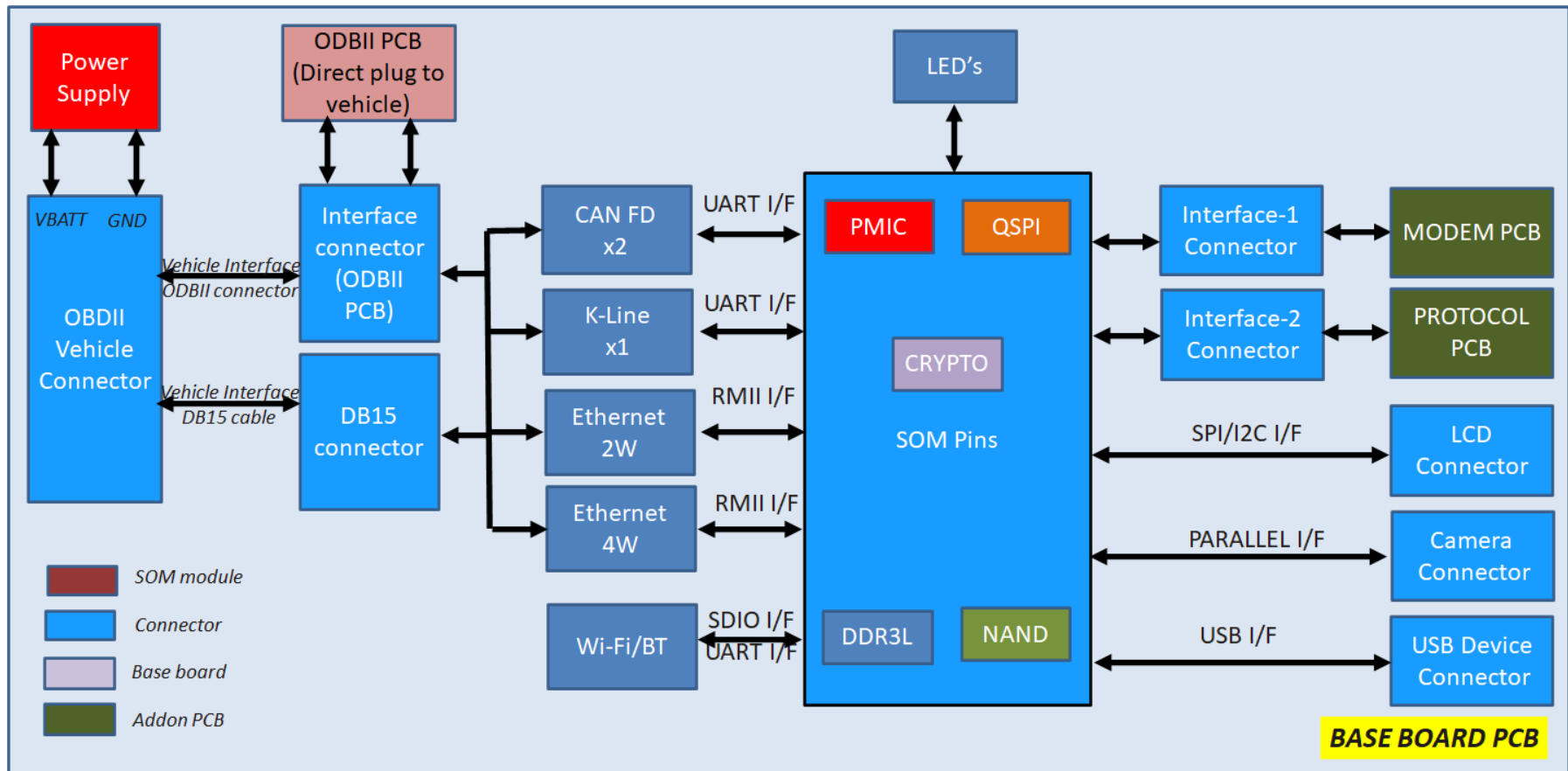
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REV.NO	DESCRIPTION	DATE
A	Alpha	28-APRIL-2023
B	Beta	01-OCTOBER-2024
C	Beta 1	12-MARCH-2025

GARUDA III V3.0 MAIN BOARD DESIGN

Schematics





PIO Muxing & Jumper setting

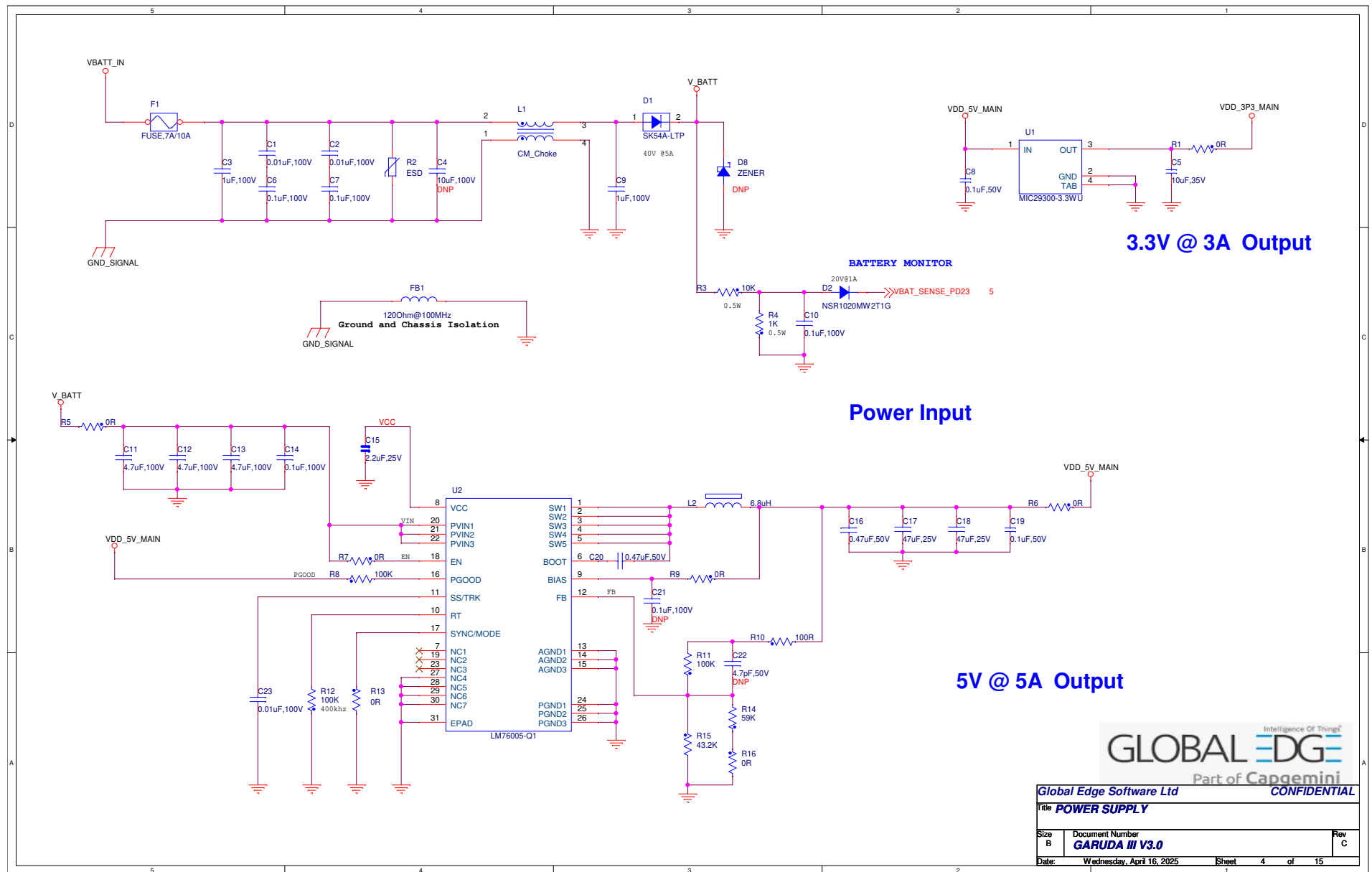
PIOA	USAGE	PIOA	USAGE	PIOB	USAGE	PIOB	USAGE	PIOC	USAGE
PA0	SDHC0_CK	PA16	SPI0_MISO	PB0	KLINE_GPIO	PB16	ETH_GRXDV	PC0	GSM_FLEXCOM0_IO4
PA1	SDHC0_CMD	PA17	SPI0_NPCS0	PB1	KLINE_GPIO	PB17	ETH_GRXER	PC1	CANTX0
PA2	SDHC0_DAT0	PA18	SDIO_DAT0	PB2	KLINE_GPIO	PB18	ETH_GRX0	PC2	CANRX0
PA3	SDHC0_DAT1	PA19	SDIO_DAT1	PB3	KLINE_URXD4	PB19	ETH_GRX1	PC3	PMIC_INT_GPIO
PA4	SDHC0_DAT2	PA20	SDIO_DAT2	PB4	KLINE_TRXD4	PB20	ETH_GTX0	PC4	ETHERNET_LED_GPIO
PA5	SDHC0_DAT3	PA21	SDIO_DAT3	PB5	QSPI1_SCK	PB21	ETH_GTX1	PC5	ERROR_LED_GPIO
PA6	SDHC0_DAT4	PA22	SDIO_CK	PB6	QSPI1_CS	PB22	ETH_GMDC	PC6	TWD1_I2C1
PA7	SDHC0_DAT5	PA23	BT_FLEXCOM2_IO1	PB7	QSPI1_IO0	PB23	ETH_GMDIO	PC7	TWCK1_I2C1
PA8	SDHC0_DAT6	PA24	BT_FLEXCOM2_IO0	PB8	QSPI1_IO1	PB24	ETH_INH_GPIO	PC8	FREE PIN
PA9	SDHC0_DAT7	PA25	BT_FLEXCOM2_IO3	PB9	QSPI1_IO2	PB25	LAN_WAKE_IN_GPIO	PC9	ETH_IRQ
PA10	SDHC0_RSTN	PA26	BT_FLEXCOM2_IO4	PB10	QSPI1_IO3	PB26	DEBUG_URXD0	PC10	FREE PIN
PA11	USB_LED_GPIO1	PA27	SPI0_NPCS2	PB11	KLINE_GPIO6	PB27	DEBUG_UTXD0	PC11	ETH ACT KEY 2
PA12	LINK_LED_GPIO2	PA28	SDIO_CMD	PB12	KLINE_GPIO	PB28	GSM_FLEXCOM0_IO0	PC12	J1708_URXD3
PA13	SDHC0_CD	PA29	SPI0_NPCS1	PB13	LAN_ENABLE_GPIO	PB29	GSM_FLEXCOM0_IO1	PC13	J1708_UTXD3
PA14	SPI0_SPCK	PA30	SDIO_CD	PB14	LAN_REFCLK	PB30	GSM_FLEXCOM0_IO2	PC14	ETH ACT KEY 1
PA15	SPI0_MOSI	PA31	SDHC0_DS	PB15	ETH_GTXEN	PB31	GSM_FLEXCOM0_IO3	PC15	WLAN_SD_INT_PC15

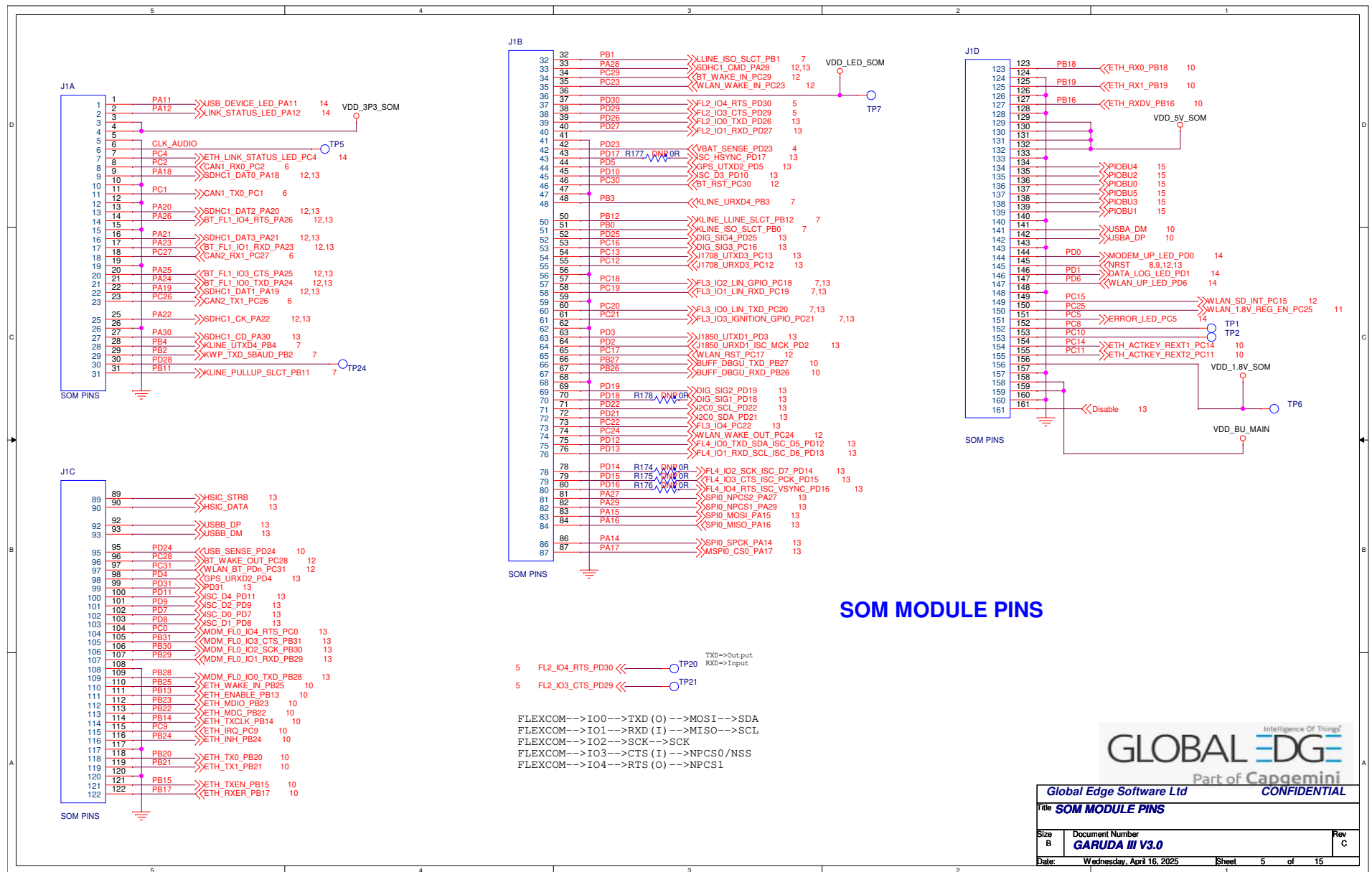
PIOC	USAGE	PIOD	USAGE	PIOD	USAGE
PC16	DIG_SIG3	PD0	MODEM LED	PD16	ISC_VSYNC/FLEXCOM4_IO4/JTAG_TDO
PC17	WLAN_RST_PC17	PD1	DATA LOG LED	PD17	ISC_HSYNC/JTAG_TMS
PC18	LIN_GPIO/FLEXCOM3_IO2	PD2	ISC_MCK/J1850_URXD1	PD18	DIG_SIG1/JTAG_NTRST
PC19	LIN_RXD_FLEXCOM3_IO1	PD3	J1850_UTXD1	PD19	DIG_SIG2
PC20	LIN_TXD/FLEXCOM3_IO0	PD4	GPS_URXD2	PD20	VMON_GPIO
PC21	FLEXCOM3_IO3/IGNITION_GPIO	PD5	GPS_UTXD2	PD21	TWD0_I2C0
PC22	FLEXCOM3_IO4	PD6	WLAN_UP_LED	PD22	TWCK0_I2C0
PC23	WLAN_WAKE_IN_PC23	PD7	ISC_D0	PD23	VBAT_SENSE
PC24	WLAN_WAKE_OUT_PC24	PD8	ISC_D1	PD24	USB_VOLTAGE_SENSE
PC25	WLAN_1.8V_REG_EN_PC25	PD9	ISC_D2	PD25	DIG_SIG4
PC26	CANTX1	PD10	ISC_D3	PD26	FLEXCOM2_IO0
PC27	CANRX1	PD11	ISC_D4	PD27	FLEXCOM2_IO1
PC28	BT_WAKE_OUT_PC28	PD12	ISC_D5/FL4_IO0_TXD_SDA	PD28	FLEXCOM2_IO2/1.8V_REG_EN
PC29	BT_WAKE_IN_PC29	PD13	ISC_D6/FL4_IO1_RXD_SCL	PD29	FLEXCOM2_IO3
PC30	BT_RST_PC30	PD14	ISC_D7/FLEXCOM4_IO2/JTAG_TCK	PD30	FLEXCOM2_IO4
PC31	WLAN_BT_PDn_PC31	PD15	ISC_PCK/FLEXCOM4_IO3/JTAG_TDI	PD31	FREE PIN

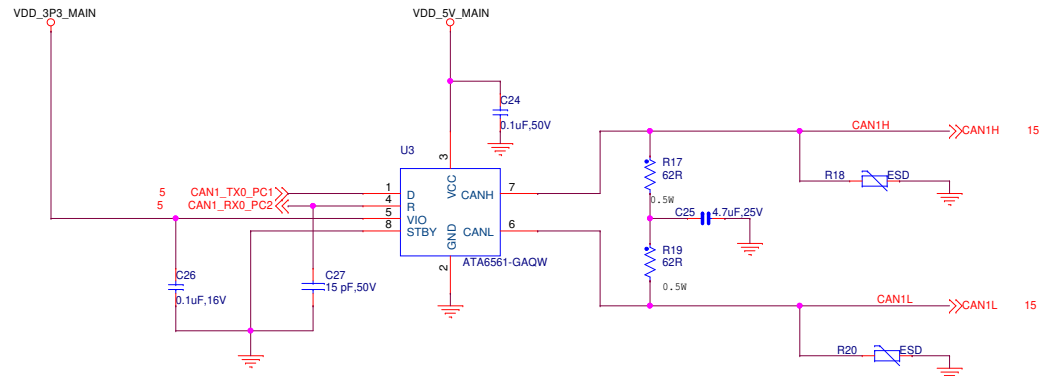
Title **PIO MUXING**

Size B Document Number **GARUDA III V3.0** Rev C

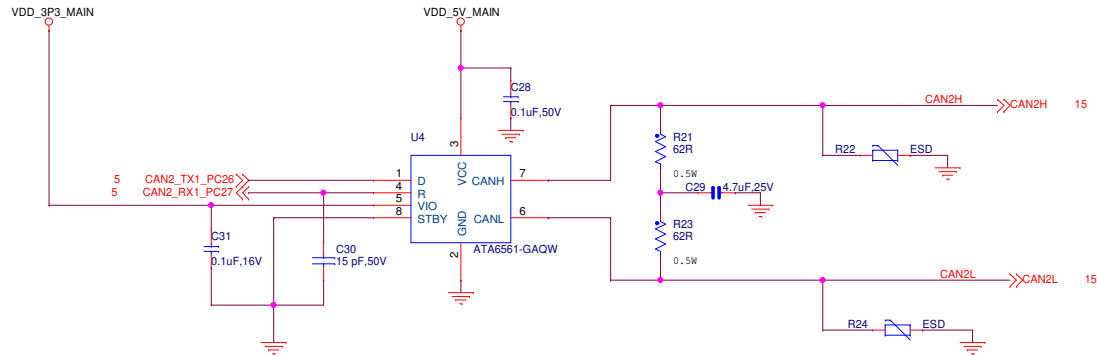
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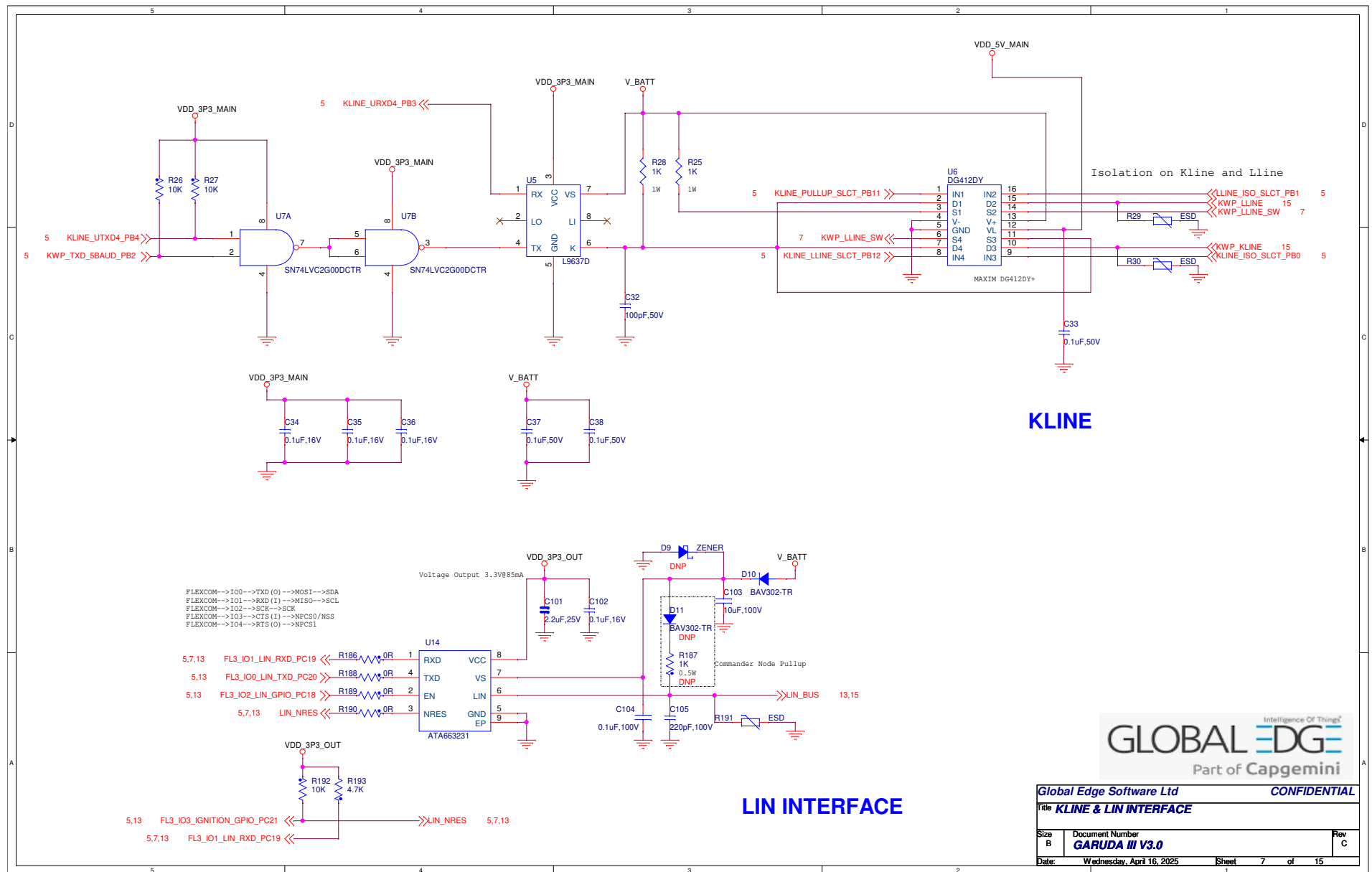
CAN 1

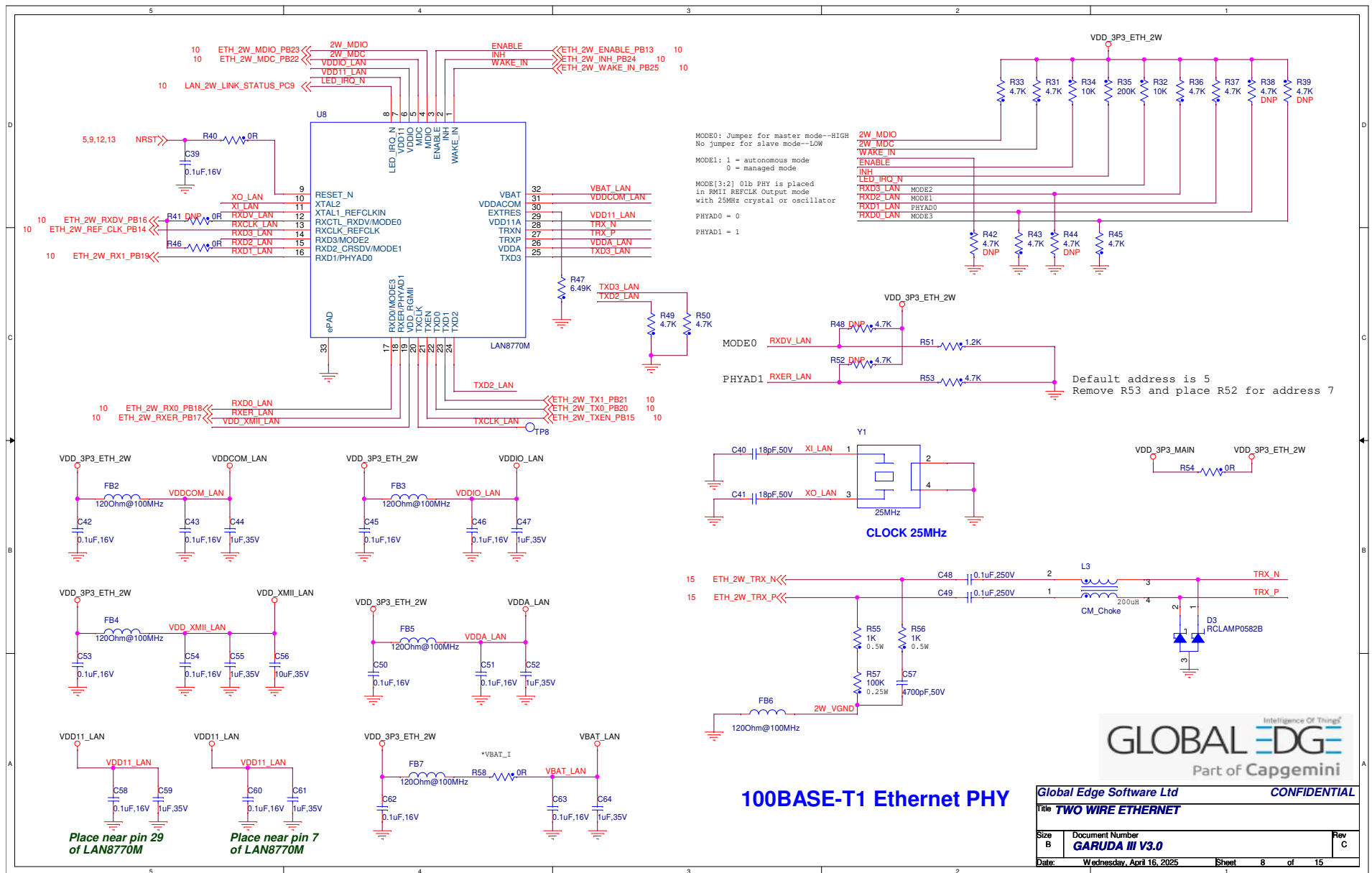


CAN 2

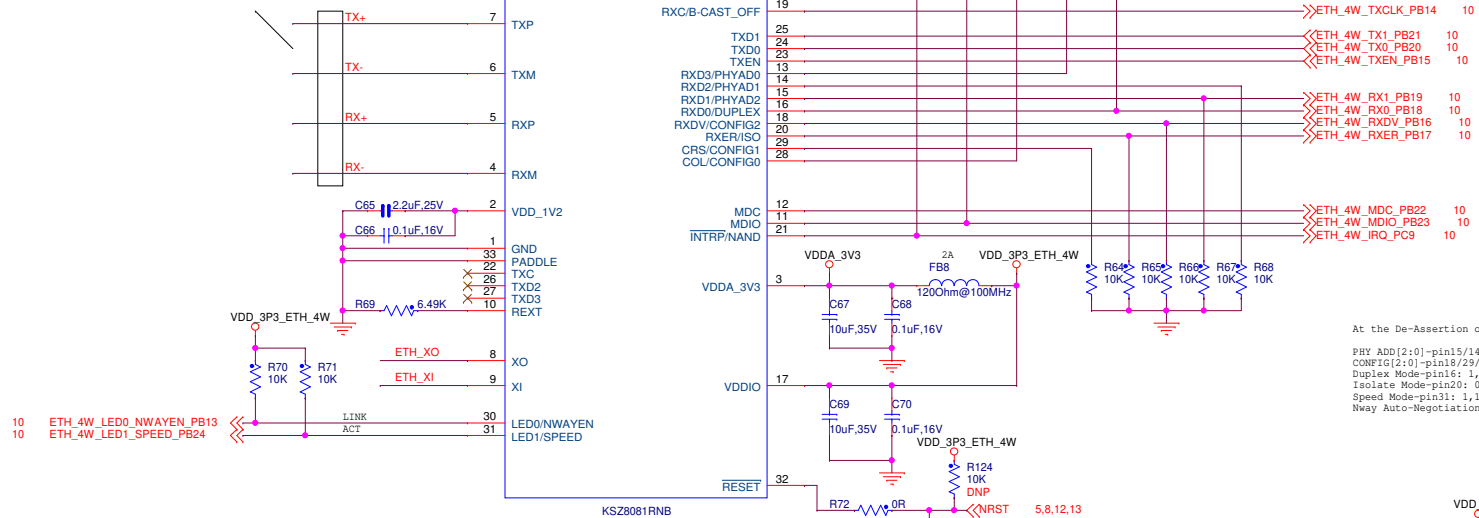


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Title CAN INTERFACE		
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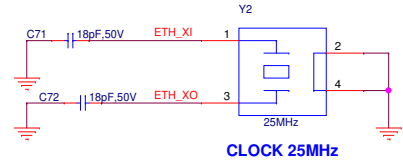
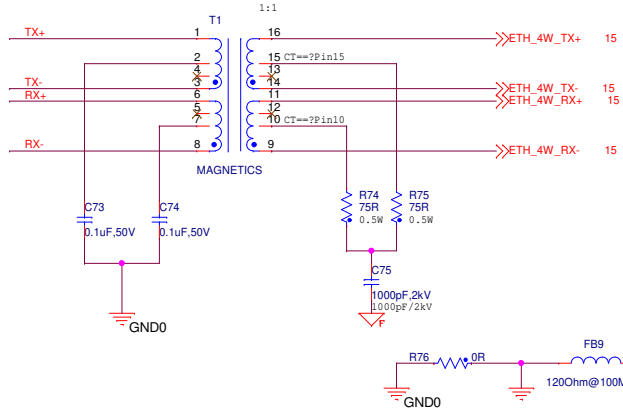
100 ohms differential trace impedance
Routing top or bottom



At the De-Assertion of Reset:

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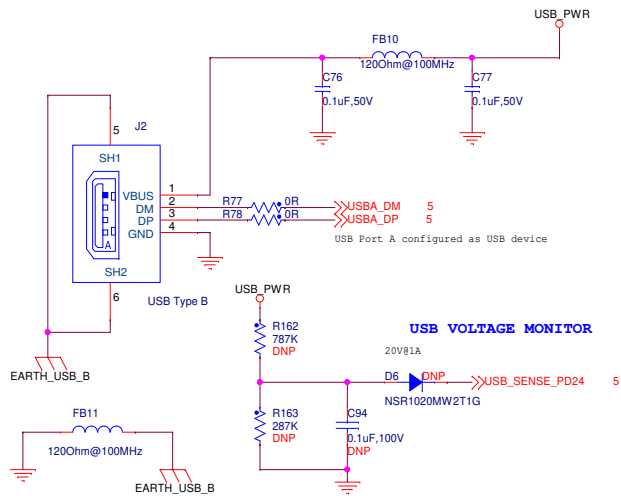
PHY ADD[2:0]-pin15/14/13: 001 = 1
CONFIG[2:0]-pin18/29/28:001,RMII mode
Duplex Mode-pin16: 1,Half Duplex
Isolate Mode-pin20: 0,Disable
Speed Mode-pin31: 1,100Mbps
Nway Auto-Negotiation-pin30: 1,Enable
  
```



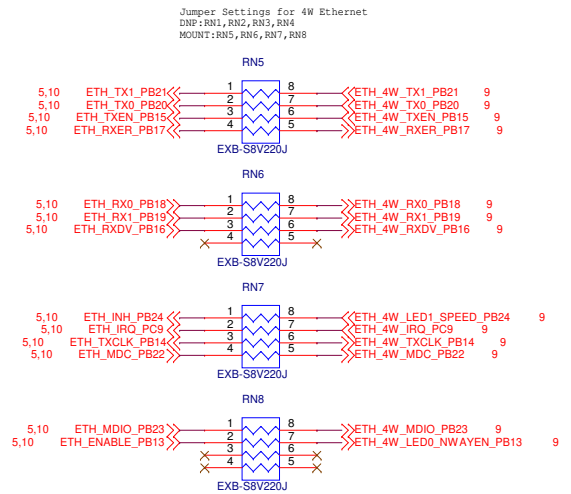
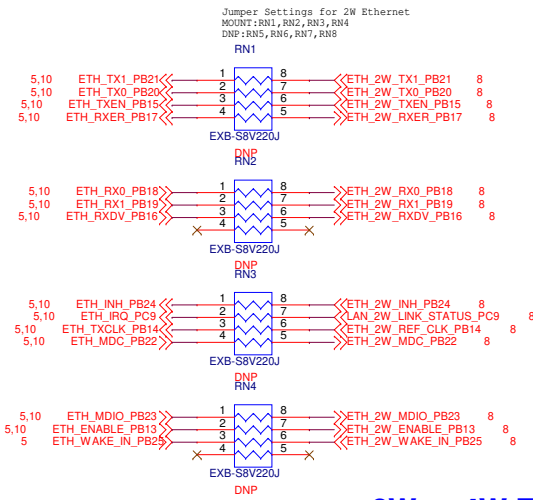
10Base-T/100Base-TX Ethernet PHY



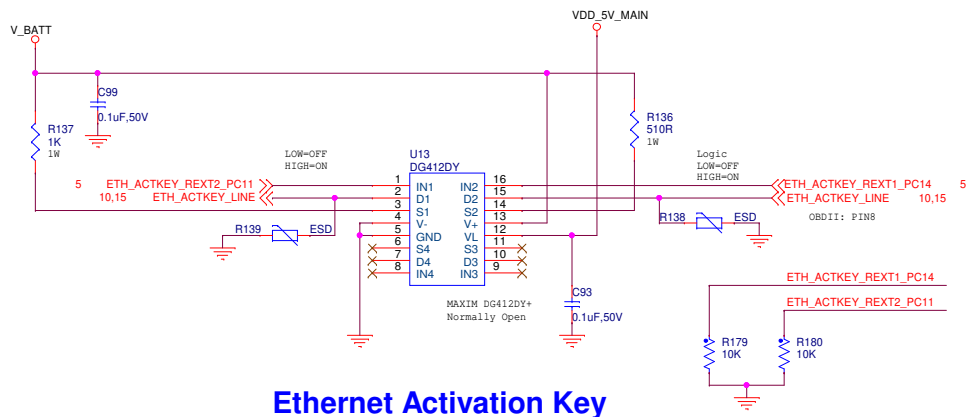
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Title FOUR WIRE ETHERNET		
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USB TYPE B

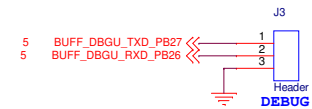


2W or 4W Ethernet Selection

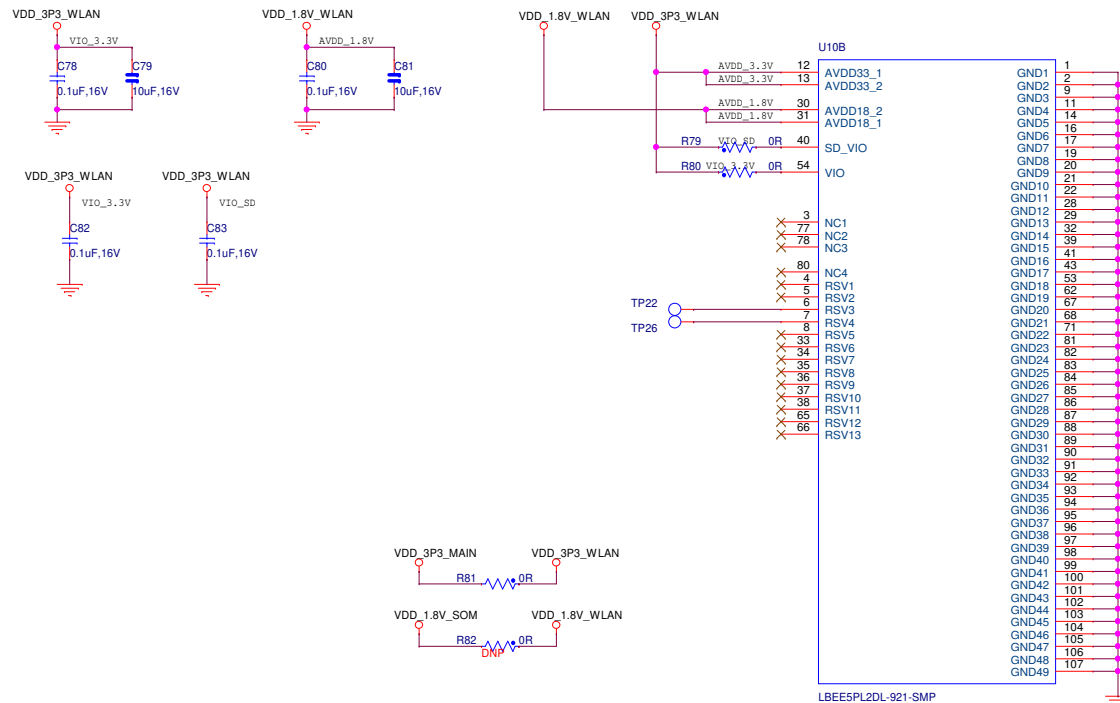


Ethernet Activation Key

BATTERY VOLTAGE	GPIO STATE	PULLUP CONFIG
Default Settings 12V	PC11==>LOW PC14==>LOW	No Pullup Open State
VBAT=12V	PC11==>LOW PC14==>HIGH	Pullup==>510ohm
VBAT=24V	PC11==>HIGH PC14==>LOW	Pullup==>1kohm
	PC11==>HIGH PC14==>HIGH	DO NOT CONFIGURE

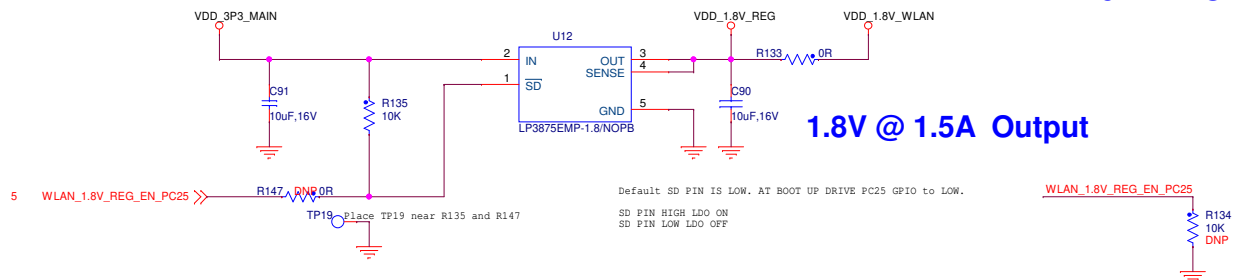


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1.8V Power Input for Wi-Fi

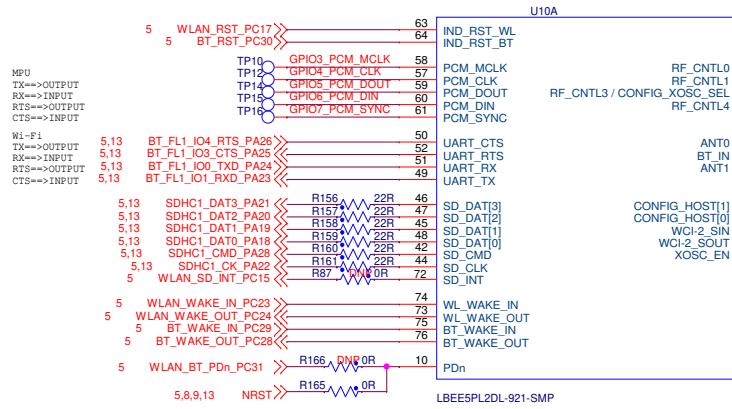
Wi-Fi & BT MURATA MODULE



1.8V @ 1.5A Output

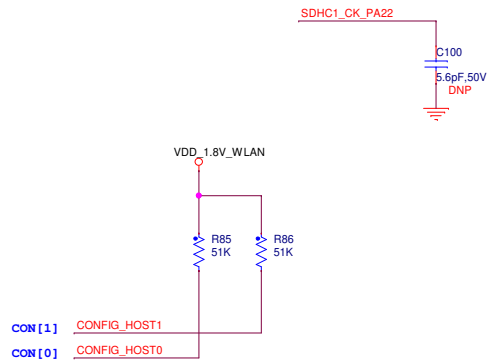
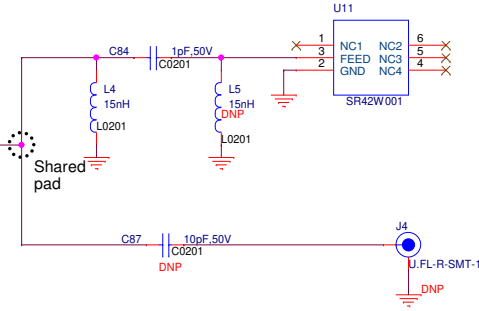


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Title Wi-Fi & BT MODULE POWER			
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Wi-Fi & BT MURATA MODULE

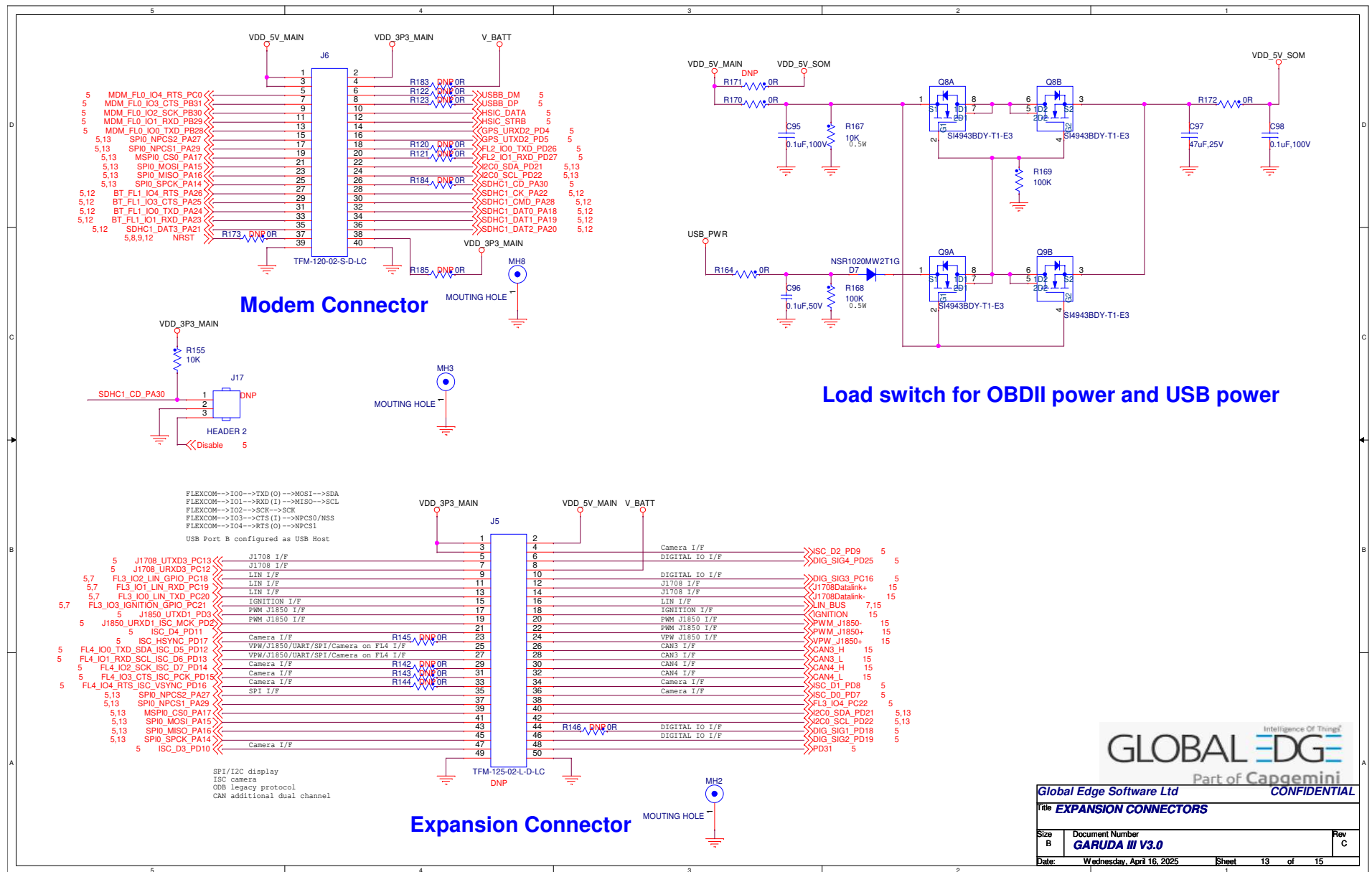
Refer datasheet for placement



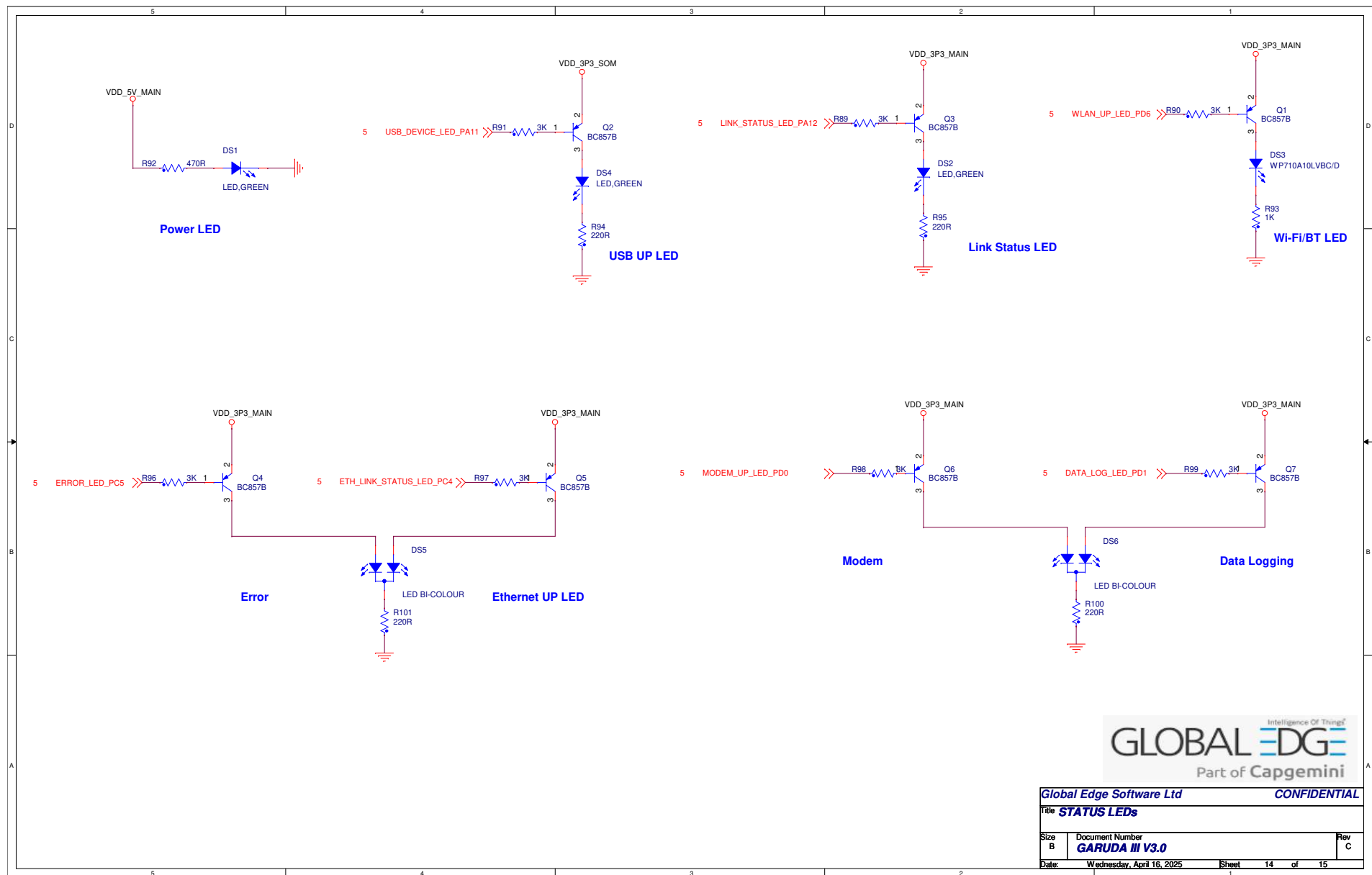
STRAP/CONFIG TABLE

CON	FUNCTION
[7]	RF_CNTL4: Reserved. Set to 1.
[5]	RF_CNTL3 / CONFIG_XOSC_SEL : Reference clock frequency select 1: 40 MHz (Default) 0: 38.4 MHz
[1:0]	CONFIG HOST BOOT[1:0]: Select the host interface used for Wi-Fi and Bluetooth
CONFIG_HOST[1:0]	Wi-Fi
11	SDIO
Others	Reserved





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Title EXPANSION CONNECTORS			
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